Hide menu

Welcome to First Principles of Computer Vision: Visual Perception

Reading: Course Syllabus 10 min

Reading: About the Instructor 10 min

Reading: Course Information and Support 10 min

Reading: Academic Honesty Policy 10 min

Reading: Discussion Forum Etiquette 5 min

Reading: Frequently Asked Questions 5 min

Pre-Course Survey

Week 1: Introduction to First Principles of Computer Vision

Course Syllabus

Welcome to First Principles of Computer Vision: Visual Perception

The ultimate goal of a computer vision system is to generate a detailed symbolic description of each image it is shown. This course focuses on the all-important problem of perception.

We first describe the problem of tracking objects in complex scenes. We look at two key challenges in this context. The first is the separation of an image into object and background using a technique called change detection. The second is the tracking of one or more objects in a video. Next, we examine the problem of segmenting an image into meaningful regions. In particular, we take a bottom-up approach where pixels with similar attributes are grouped together to obtain a region.

Finally, we tackle the problem of object recognition. We describe two approaches to the problem. The first directly recognizes an object and its pose using the appearance of the object. This method is based on the concept of dimension reduction, which is achieved using principal component analysis. The second approach is to use a neural network to solve the recognition problem as one of learning a mapping from the input (image) to the output (object class, object identity, activity, etc.). We describe how a neural network is constructed and how it is trained using the backpropagation algorithm.

Course Prerequisites

Learners are required to know the fundamentals of linear algebra and the fundamentals of calculus. No prior knowledge of imaging or computer vision is assumed. The knowledge of any programming language would enable the learners to understand how the methods described in the course can be implemented in software. However, the knowledge of a programming language is not required as the course does not have programming assignments.

Learning Objectives

The ultimate goal of any computer vision system is to perform some form of perception. The methods described in this course, therefore, have implications for virtually all fields that use computer vision as a technology. This course is designed to achieve the following learning goals:

- Design algorithms for detecting meaningful changes in a scene.
- Develop methods for tracking objects in a video while the object undergoes changes in pose and illumination, and may at times be occluded by other objects.
- Learn several approaches to segmenting an image into meaningful regions.
- Create an end-to-end pipeline for learning and recognizing objects based on their visual appearance.
- Learn the fundamentals of neural networks and the backpropagation algorithm used to train networks.

Course Schedule

This is a graduate-level course, with an estimated 4-6 hours of study time per week.

Grading Policy

- Quizzes count towards 100% of the grade. You will be given 2 attempts for each quiz every 8 hours.
- **A passing grade for the course is 70% or higher.**

Course Outline

Week 1: Introduction

1.1 Overview of Introduction
1.2 What is Computer Vision?
1.3 What is Vision Used For?
1.4 How Do Humans Do It?
1.5 Topics Covered
1.6 About the Lecture Series
1.7 References and Credits

Week 2: Object Tracking

2.1 Overview of Object Tracking
2.2 Change Detection
2.3 Gaussian Mixture Model
2.4 Object Tracking using Template Matching
2.5 Tracking by Feature Detection

Week 3: Image Segmentation

3.1 Overview of Image Segmentation
3.2 Segmentation in Humans
3.3 Segmentation as Clustering
3.4 k-Means Segmentation
3.5 Mean Shift Segmentation
3.6 Graph Based Segmentation

Week 4: Appearance Matching

4.1 Overview of Appearance Matching
4.2 Shape vs. Appearance
4.3 Learning Appearance
4.4 Principal Component Analysis
4.5 Finding Principal Components
4.6 PCA and SVD
4.7 Parametric Appearance Representation
4.8 Appearance Matching

Week 5: Neural Networks

5.1 Overview of Neural Networks
5.2 Perceptron
5.3 Perceptron Network
5.4 Activation Function
5.5 Neural Network
5.6 Gradient Descent
5.7 Backpropagation Algorithm
5.8 Example Applications
5.9 When to use Machine Learning?

Recommended Readings

Textbooks

- Computer Vision: Algorithms and Applications, Szeliski, R., Springer.
- Computer Vision: A Modern Approach, Forsyth, D and Ponce, J., Prentice Hall.
- Robot Vision, Horn, B. K. P., MIT Press.
- A Guided Tour of Computer Vision, Nalwa, V., Addison-Wesley.
- Digital Image Processing, González, R and Woods, R., Prentice Hall.
- Optics, Hecht, E., Addison-Wesley.
- Eye and Brain, Gregory, R., Princeton University Press.
- Animal Eyes, Land, M. and Nilsson, D., Oxford University Press.

Papers

In addition to the above textbooks, the course also presents concepts that are described in published research papers. The papers are listed at the end of the last video clip of each topic –object tracking, image segmentation, appearance matching and neural networks.

Go to next item Completed

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